

HYDROFARMS AGRI SOLUTIONS

SURVEY, BENCHMARKS AND SETTING OUT

Activity-Specific Construction Method Statement

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Project	One-Feddan NFT Lettuce Greenhouse Project
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Prepared by	Hydrofarms Agri Solutions
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CONTROLLED FIELD PROCEDURE
No excavation, foundation, buried service, structural erection or equipment placement shall start from an unverified dimension. All setting out shall be derived from the approved survey control network and the latest coordinated AFC drawings.

Document Control

Approval and responsibility

Role	Primary action
Project manager	Authorises issue and ensures resources and interfaces are available.
Discipline engineer	Provides coordinated AFC geometry, grids, levels and setting-out requirements.
Lead surveyor	Establishes control, performs observations, calculations, checks and records.
Independent checker	Repeats critical observations and confirms closure before release.
QA/QC engineer	Controls calibration, ITP intervention, inspection requests and survey records.
Construction manager	Protects control points and prevents work from unapproved marks.
Client/consultant	Witnesses or approves designated hold points in accordance with the ITP.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method defines how the project survey control network, benchmarks, grids, construction set-out points and as-built records will be established, checked, protected and transferred between civil, structural, hydraulic, mechanical and electrical work packages.

It establishes a repeatable process that prevents cumulative dimensional error across the 126 m production length, the 36 m greenhouse width, the western service greenhouse and the southern tank/WTP/equipment zone.

2. Scope

- Site reconnaissance, boundary confirmation and existing-condition survey.
- Establishment of permanent horizontal control points and temporary benchmarks.
- Definition of the project local coordinate grid and elevation datum.
- Setting out the production greenhouse, service greenhouse and external utility/equipment areas.
- Setting out excavations, tank rooms, foundations, pedestals, anchors, sleeves and buried networks.
- Control surveys during structural erection and equipment installation.
- As-built surveying of concealed and completed works.
- Survey record control, checking, acceptance and issue of setting-out releases.

3. Governing Information

Priority	Controlled information
1	Approved project coordinate/datum information and statutory boundary data
2	Latest coordinated AFC site plan, civil drawings and structural grid drawings
3	Approved tank-room, pipe-route, drainage, equipment and builders-work drawings
4	Approved shop drawings, anchor templates and manufacturer setting-out requirements
5	HF-GH01-MST-GEN-001 and the approved activity ITP / risk assessment

DRAWING STATUS CHECK

The surveyor shall record the drawing number, revision and issue status on every set-out sheet. Superseded or uncontrolled drawings shall not be used, even when their dimensions appear unchanged.

4. Project Survey Basis

Element	Setting-out basis
Production greenhouse	Nominal 126 m east-west length by 36 m north-south width
Longitudinal structural grids	Fourteen nominal 9 m bays producing fifteen principal frame lines
Transverse structural grids	Nine nominal 4 m spans across 36 m, subject to the latest AFC structural grid
Service greenhouse	9 m x 36 m, located on the western side of the production greenhouse
Vertical control	5.5 m underside/eaves reference and 7.0 m highest arch point, verified to structural drawings
Central operating aisle	2.5 m clear corridor through the production greenhouse
Inter-table aisles	0.75 m clear operating aisles
Southern external zone	Production/nursery nutrient tanks, reserve tanks, chillers and WTP aligned along the southern side as shown on AFC drawings
Buried services	Production feed, return, drainage, WTP and auxiliary networks set out by centreline and invert/elevation

Recommended local coordinate convention

Unless the approved CAD survey file already defines another convention, the project local grid shall use the X-axis along the 126 m production length, the Y-axis across the 36 m width and Z for elevation. The origin shall be a protected approved control point outside construction disturbance. Absolute coordinates and datum shall be transferred from the client/authority control and recorded in the Survey Control Point Register.

This convention is a calculation and communication aid only. The signed approved control schedule governs the site.

5. Personnel and Equipment

Minimum survey team

- One lead surveyor competent in total-station observations, levelling, coordinate calculations and CAD data.
- One survey assistant for prism, levelling staff, marking and point protection.
- One independent checker for primary control, critical grids, anchor templates and as-built closure.
- QA/QC attendance at hold points and concealed-work surveys.

Survey instruments

Instrument / item	Minimum requirement	Use
Total station	Angular accuracy 5 seconds or better; distance accuracy suitable for project control	Horizontal control, grids and coordinates
Automatic/digital level	Calibrated; staff and tripod in good condition	Benchmarks, formation, slabs and inverts
GNSS receiver	Use only where site control and sky visibility support the required accuracy	Transfer/reference; not sole basis for anchors
Steel tape	Calibrated 30/50 m tape	Independent short-distance and diagonal checks
Prisms/targets	Matched constants; clean and undamaged	Total-station observations
Control-point hardware	Stainless pin/bolt or concrete monument with durable identification	Permanent reference network
Marking materials	Paint, nails, witness stakes, profile boards and protective barriers	Construction transfer and protection
Data tools	Approved CAD/base file, field controller and controlled calculation sheets	Coordinate preparation and records

Calibration and checks

- Calibration certificates shall be valid and traceable on the day of use.
- Perform instrument collimation, compensator, prism-constant and level two-peg checks at mobilisation and at defined intervals.
- Repeat checks after transport damage, suspected movement, extreme weather exposure or unexplained closure error.

- Record instrument serial number and calibration status on every survey report.

6. Preconditions

- Approved boundary/site information and access to the control reference are available.
- Latest coordinated AFC drawings and a controlled digital CAD base are issued.
- The site is safe to access and known utilities or hazards are identified.
- Primary control-point locations are outside excavation, traffic, storage and structural movement zones.
- The surveyor has reviewed cumulative dimensions, diagonals, offsets and cross-discipline interfaces.
- The inspection notice period and consultant witness requirements are confirmed.

7. Detailed Method

7.1 Site reconnaissance and existing-condition survey

- Walk the complete site with the construction and HSE teams; identify boundaries, access, obstructions, overhead/underground services, drainage paths and unstable ground.
- Locate the client/authority reference points and verify that point descriptions correspond with physical monuments.
- Survey existing ground levels on a grid adequate for earthwork and drainage design verification, with additional points at changes of slope and proposed tank rooms.
- Issue an existing-condition plan with photographs and report any conflict with the design footprint before setting out.

7.2 Establish primary horizontal control

- Install a minimum of three well-distributed permanent control monuments so that at least two remain intervisible and any point can be re-established from independent observations.
- Observe the control as a closed traverse or equivalent redundant network tied to the approved reference.
- Measure both faces/sets where appropriate and balance the network only after raw closure has been reviewed.
- Record original observations, instrument/prism data, calculated closure, adjustment and final coordinates.
- Independently reoccupy or check the network before it is released for construction.

7.3 Establish vertical control

- Transfer datum by closed differential levelling from the approved reference.
- Install at least two protected temporary benchmarks on opposite sides of the work so construction can continue if one point is disturbed.
- Close the levelling run to the origin or an independently verified benchmark and calculate misclosure before adjustment.
- Clearly mark benchmark identification and reduced level; do not use movable kerbs, temporary steel or tank covers as benchmarks.

7.4 Establish the greenhouse grid

- Set the four production-greenhouse corners from control coordinates, not by chained dimensions from site fences.
- Independently check width, length, both diagonals and orientation before placing intermediate grid lines.
- Set the fifteen principal frame lines along the 126 m length from calculated coordinates. Avoid cumulative tape stepping.
- Set transverse column/arch grids from calculated coordinates and confirm the latest structural spacing before marking.
- Place offset witness points/profile boards outside excavation and protect them with durable labels.
- Set the service greenhouse on the western side from the same primary network so the shared wall, central corridor, doors and external hygiene route align correctly.

7.5 Set out southern tanks, WTP and equipment zone

- Use the coordinated external-services drawing to set the aligned production tanks, reserve-tank room, nursery tank room, chillers and WTP along the southern side.
- Mark excavation limits, structure centreline, finished floor/cover level, access openings, pipe sleeves and maintenance/removal clearances separately.
- Confirm that tank-room and equipment positions do not obstruct fan airflow, doors, service routes or future maintenance lifting paths.
- Issue a coordinated setting-out release signed by civil, mechanical and electrical engineers before excavation.

7.6 Foundations, pedestals and anchors

- Set footing/pedestal centre points and excavation extents from the grid; transfer offsets outside excavation.
- Check blinding level, formwork centre, pedestal top level, sleeve locations and anchor template orientation before concrete.
- Survey anchor bolts in X, Y, elevation, projection and spacing while the template remains adjustable, then recheck immediately before pour.
- Perform an as-cast anchor survey after form removal and before structural steel delivery/erection.
- Do not force columns onto out-of-tolerance bolts; issue an NCR and obtain an engineered disposition.

7.7 Buried pipes, drainage and sleeves

- Set pipe centreline, trench width, chamber locations, crossings and design invert levels from the approved coordinated route drawings.
- Use offset stakes so line and level can be recovered after excavation.
- Check pipe route and invert at direction changes, tees, chambers, tank entries, crossings and regular intervals sufficient to demonstrate continuous fall.
- After testing but before covering, survey the installed pipe centreline/invert, fittings, valves, cleanouts, sleeves and chamber covers.
- Upload the as-built points to the controlled CAD model and obtain concealed-work release before bedding surround/backfill closes the work.

7.8 Structural erection control

- Verify base plate level and grid before erection.
- Check column plumbness and frame alignment after temporary bracing, after permanent bracing and before envelope installation.
- Check eaves/underside reference at 5.5 m and crown/high-point reference at 7.0 m against the approved structural geometry.
- Check roof-vent, gutter, fan-frame and cooling-pad-frame interfaces before cladding conceals adjustment points.
- Issue a structural survey report with measured deviations and a signed release before covering and screen systems proceed.

7.9 Final and as-built survey

- Survey all permanent works required by the turnover index, including greenhouse grids, tank rooms, equipment bases, drainage, buried networks and access levels.
- Use the same coordinate/datum system as construction control and retain raw observation files.
- Show design versus as-built values and highlight accepted deviations; do not silently alter design geometry in the as-built drawing.
- Issue signed PDF survey reports and editable CAD/coordinate files under document control.

8. Acceptance Tolerances

The following project tolerances apply unless a stricter AFC drawing, structural calculation, manufacturer requirement or contract specification governs. Tolerances are not cumulative and do not permit reducing required clear widths.

Item	Check	Acceptance criterion
Primary horizontal control	Network closure and adjustment	Demonstrate survey-grade closure; final control coordinate repeatability within ± 3 mm
Vertical control	Closed levelling run	Benchmark repeatability within ± 2 mm
Production greenhouse corners	Plan position	± 5 mm from approved coordinates
Production footprint	Length / width	± 5 mm per principal dimension; no cumulative stepping
Footprint diagonals	Difference from calculated diagonals	Maximum 10 mm unless stricter structural criterion applies
Principal grid line	Plan offset	± 3 mm
Foundation/pedestal centre	Plan position	± 5 mm
Pedestal/base level	Elevation	± 3 mm
Anchor group	Centre / orientation	± 3 mm plan; orientation to template detail
Individual anchor spacing	Bolt-to-bolt	± 2 mm or stricter steel-fabricator template requirement

Item	Check	Acceptance criterion
Anchor projection	Vertical projection	±3 mm
Column plumbness	Completed frame	Within approved structural tolerance; target not greater than H/1000
Buried pressure pipe	Horizontal route	±20 mm unless sleeve/equipment connection requires tighter control
Gravity pipe invert	Elevation	±5 mm with continuous design fall and no reverse grade
Tank/equipment base	Plan / level	±5 mm plan and ±3 mm level unless vendor requires tighter tolerance
Finished floor/drain level	Elevation/fall	±5 mm while maintaining positive drainage and required thresholds

TOLERANCE ESCALATION

Any result outside tolerance shall be reported before dependent work starts. The surveyor may not average, relocate or remark a failed point without an approved engineering disposition and revised controlled record.

9. Inspection and Test Plan

No.	Inspection stage	Method	Type	Responsibility	Record
01	Approved survey reference and AFC drawings	Document review	R	QA/QC / Surveyor	Document-control check
02	Instrument calibration and field checks	Certificate + test	H	QA/QC / Surveyor	Calibration register
03	Existing-condition survey	Field survey	W	Surveyor / Consultant	Existing-condition plan
04	Primary control network	Closed observations and calculation	H	Independent checker / Consultant	Control-network report
05	Temporary benchmarks	Closed levelling	H	Independent checker	Benchmark register
06	Production/service greenhouse corners	Coordinates, dimensions and diagonals	H	QA/QC / Consultant	Grid acceptance sheet
07	Southern equipment/tank zone	Coordinated setting out	H	Civil/MEP/QA-QC	Interface release
08	Excavation and formation	Plan and level	W	Civil QA/QC	Formation survey
09	Formwork, sleeves and anchors	Pre-pour survey	H	Structural QA/QC / Consultant	Pre-pour survey sheet
10	As-cast anchors	Coordinate and level survey	H	Structural engineer	Anchor as-built report
11	Buried services before covering	Route/invert/as-built survey	H	MEP QA/QC / Consultant	Concealed-work survey
12	Structural frame alignment	Grid, plumbness and level	H	Structural engineer / Consultant	Frame survey report
13	Final as-built	Independent review and CAD update	R	QA/QC / Client	As-built survey dossier

Intervention codes: H = Hold, W = Witness, R = Record review. Notice periods shall follow the project quality plan.

10. Hold Points

- HP-SUR-01: Approval of instrument calibration and the primary control network before construction setting out.
- HP-SUR-02: Acceptance of production and service greenhouse corners, dimensions and diagonals before excavation/foundations.
- HP-SUR-03: Coordinated release of the southern tank/WTP/equipment zone before excavation.
- HP-SUR-04: Acceptance of anchor templates, sleeves and pedestal levels before concrete.
- HP-SUR-05: Acceptance of buried-service as-built route and invert before covering.
- HP-SUR-06: Acceptance of the erected structural frame survey before envelope installation.

11. HSE and Environmental Controls

Hazard	Potential consequence	Mandatory control	Residual risk
Traffic / mobile plant	Surveyor struck while occupying points	Survey exclusion zone, banksman, high-visibility PPE and coordinated work-front permit	Medium
Excavation edges	Fall or collapse exposure	No entry without excavation release; maintain safe offset and protected access	Medium
Work at height	Checking columns/arches/vents	Use approved MEWP/platform; no climbing on unapproved frame	Medium
Laser/optical equipment	Eye exposure / distraction	Use Class-compliant equipment, controlled aiming and trained operators	Low
Heat and sun	Heat stress and reduced concentration	Hydration, shade, work/rest cycle and rescheduling during peak heat	Medium
Manual handling	Tripod, batteries, monuments	Team lift where needed, correct carrying cases and clear routes	Low
Night/low light	Trip and reading errors	Provide task lighting; do not set critical control under inadequate visibility	Low
Control-point installation	Dust, drilling and hand-tool injury	Eye/hand/hearing protection, dust control and tool inspection	Low

Environmental controls

- Use durable monuments with the minimum necessary excavation and reinstate disturbed surfaces.
- Do not mark permanent finished surfaces with non-removable paint unless approved.
- Collect used batteries, paint cans, stakes and packaging; do not leave survey waste in drainage or crop zones.
- Protect existing trees, services, drainage routes and neighbouring property during point establishment.

12. Survey Records and Deliverables

Record	Minimum content	Issue timing
Survey Control Point Register	Point ID, description, coordinates, RL, monument type, photograph, date, status	Before construction setting out
Instrument Register	Instrument/serial, calibration certificate, field checks, due date	Current throughout works
Existing-Condition Survey	Contours/spot levels, boundaries, obstructions, utilities and photographs	Before earthworks
Setting-Out Sheet	Drawing revision, point list, coordinates/levels, offsets, observations and checker	Before each work-front release
Grid Acceptance Sheet	Corners, dimensions, diagonals, frame grids and service greenhouse interface	Before excavation/foundations
Pre-Pour Survey	Formwork, centres, sleeves, anchors, levels and deviations	Before concrete
Concealed-Work Survey	Pipe route/invert, fittings, valves, chambers, sleeves and photographs	Before covering
Structural Alignment Report	Base/grid, plumbness, eaves, crown, bracing and interface points	Before envelope
Final As-Built Survey	Coordinates, levels, design comparison, signed CAD/PDF and raw data index	Before turnover

File naming

SURVEY RECORD CODE
HF-GH01-SUR-[AREA]-[TYPE]-[NNN]-R[REV] AREA identifies production, service, southern utilities or external works. TYPE identifies control, set-out, pre-pour, concealed or as-built survey.

Daily close-out

- Download and back up raw observations before leaving the project data system.
- Record the day's instrument, weather, drawing revision, control points used and points established.
- Flag disturbed/missing control points and suspend dependent set-out until re-established.
- Issue signed set-out/inspection sheets through document control; field marks alone are not an acceptance record.

Appendix A - Survey Control Point Register Template

Point ID	Easting X	Northing Y	RL Z	Description / monument	Check date	Status
CP-01	Entered from approved survey	Entered from approved survey	Entered from approved datum	Permanent protected monument	Initial and periodic	Active / replaced
CP-02	Entered from approved survey	Entered from approved survey	Entered from approved datum	Permanent protected monument	Initial and periodic	Active / replaced
TBM-01	Reference coordinate	Reference coordinate	Approved reduced level	Protected benchmark	Before level transfer	Active / replaced

Actual coordinates shall be populated only from the signed project control survey and shall not be inferred from nominal greenhouse dimensions.

Appendix B - Setting-Out Release Checklist

- Latest AFC drawing number and revision recorded.
- Control points and benchmark verified before use.
- Independent coordinate calculation completed.
- Corners, dimensions, diagonals and offsets checked.
- Interfaces with adjacent discipline confirmed.
- Marks are durable, labelled and protected.
- Measured deviations are within acceptance criteria.
- Raw data, calculation sheet and sketch attached.
- QA/QC and required witness/hold-point signatures obtained.

NEXT METHOD STATEMENT

After approval of the survey control network and setting-out process, issue HF-GH01-MST-CIV-001 for excavation, dewatering, backfilling and compaction. Excavation shall not commence until HP-SUR-02 and HP-SUR-03 are released.